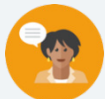
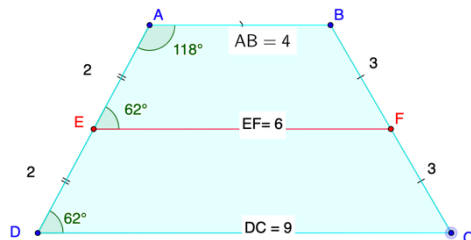


8.7 Trapezoid Midsegment Theorem

Lesson Introduction

 Benchmark	MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.		 Learning Targets	- I can prove the trapezoid midsegment theorem. - I can use properties of trapezoids to solve mathematical and real-world problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the trapezoid midsegment theorem? - How can midsegments be used to find missing lengths of a trapezoid?			
 Lesson Narrative	In this lesson, students will continue their work with trapezoids to discover the Trapezoid Midsegment Theorem. Once students have an opportunity to explore and construct a definition for the Trapezoid Midsegment Theorem, they are given an opportunity to apply the concept theoretically, and in real-world situations.			
 Component Summary	8.7.1 Warm-Up (~5 min.)	Notice and Wonder for trapezoids in architecture. (SE p. 43)	8.7.4 Guided Practice (10 min.)	Find measures using the trapezoid midsegment theorem. (SE p. 46)
	8.7.2 Exploration (5-10 min.)	Explore the relationship between lengths in a trapezoid. (SE p. 43)	8.7.5 Your Turn! (5-10 min.)	Find measures using the trapezoid midsegment theorem. (SE p. 47)
	8.7.3 Guided Instruction (5-10 min.)	Prove the trapezoid midsegment theorem. (SE p. 44)	8.7.6 Wrap-Up (~5 min.)	Summarize the lesson as a note to a friend. (SE p. 48)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Midsegment - Trapezoid Midsegment Theorem <u>Additional Glossary:</u> N/A		- Patty paper (Optional) Scissors	
			<u>Additional Preparation</u> N/A	

 Teacher Tips	Common Misconceptions Students may set half the length of the midsegment and one of the bases equal to the other base.	Things to Consider Students will continue the work of the triangle midsegment theorem in a later lesson. For this lesson, students just need a surface understanding of what the midsegment of a triangle is, and how its length is related to the base of the triangle.
	Literacy and Language Routines Notice and Wonder This routine provides a low-floor, high-ceiling approach to create pathways that connect concrete observations to abstract concepts. Each activity in this lesson can use Notice and Wonder, as students can connect what they see about midsegments to their mathematical properties, ultimately formalizing this connection as the Trapezoid Midsegment Theorem.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure In the 8.7.2 Exploration activity, students make use of measurements to begin formulating the patterns and connections that ultimately become their working definition of the Trapezoid Midsegment Theorem. Utilize this opportunity to build students' capacity for making use of patterns, rather than memorizing a theorem.
 Differentiate Instruction	Digital applets provide manipulatives for students to explore midsegments without having to constantly erase and draw new shapes. For example, GeoGebra has a trapezoid midsegment applet in which students can adjust the length of the base to predict and verify how the length of the midsegment might change. Consider utilizing digital manipulatives during 8.7.4 Guided Practice or 8.7.5 Your Turn! for additional scaffolding or extending to predict lengths for other shapes.	
Lesson Notes:		

8.7.1 Warm-Up: Trapezoids in Architecture

Component Overview: Students engage in a Notice and Wonder routine to make meaningful connections to trapezoids in the real world.



8.7.1 Warm-Up: Trapezoids in Architecture

Four different buildings are shown.

Machu Picchu, Peru	Greenland Center, Beijing, China
	
Garden Terrace, Nagasaki	International Spy Museum, Washington D.C.
	



Notice and Wonder

Allow students the opportunity to notice and wonder individually and then discuss with a partner the possible reasons the trapezoids were used the way they were.

- What do you notice about the buildings?
Answers will vary. I notice that they all include a trapezoid shape.
- Describe how a trapezoid is used in each building.
Answers will vary. In Machu Picchu, a trapezoid is used as an entryway. The Greenland Center uses inverted trapezoids similar to bricks. Garden Terrace appears to be in the shape of a trapezoid. The International Spy Museum uses trapezoid shapes at one end of the building.

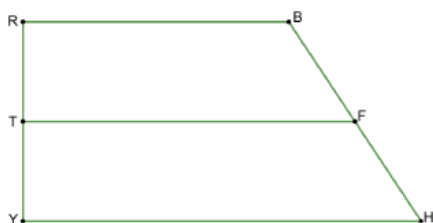
8.7.2 Exploration: Midsegment of a Trapezoid

Component Overview: Students use patty paper to explore visual and mathematical relationships between trapezoids and their midsegments. Students can work in pairs or small groups as the teacher facilitates.



8.7.2 Exploration: Midsegment of a Trapezoid

1. Trapezoid $RBHY$ is shown, where $\overline{RB} \parallel \overline{YH}$.



- a. Use patty paper or a ruler to determine if $\overline{TF}$ is the midsegment of trapezoid $RBHY$. Justify your conclusion.

Point T is the midpoint of $\overline{RY}$ and point F is the midpoint of $\overline{BH}$, therefore $\overline{TF}$ is the midsegment.

- b. Use a ruler to complete the table.

Answers will vary depending on the units used.

Segment(s)	Measure
$\overline{RB}$	$RB = 6.7 \text{ cm}$
$\overline{YH}$	$YH = 10 \text{ cm}$
$\overline{TF}$	$TF = 8.35 \text{ cm}$

2. What do you notice about the relationship between the length of the midsegment and the sum of the lengths of the bases?

The length of the midsegment is equal to half the length of the sum of the bases.

3. Write an equation to show the relationship between the length of the midsegment of a trapezoid and the length of its bases.

$$\text{Midsegment} = \frac{1}{2}(b_1 + b_2)$$



Allow students to choose any unit of measure for this activity. The units themselves do not matter, but the relationship between the lengths does.



Notice and Wonder

Consider implementing this routine to allow students an opportunity to generate ideas about trapezoids. Monitor for student responses that can later be revisited and formalized to make meaningful connections during the proof in 8.7.3 Guided Instruction.

8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

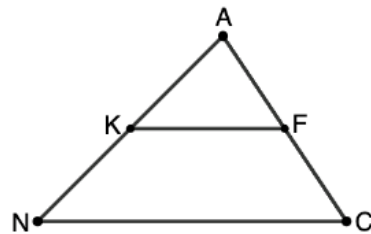
Component Overview: Teacher facilitates a proof that connects different properties of a trapezoid to the Trapezoid Midsegment Theorem.



8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

The **midsegment** of a triangle is a segment joining the midpoints of two sides of a triangle.

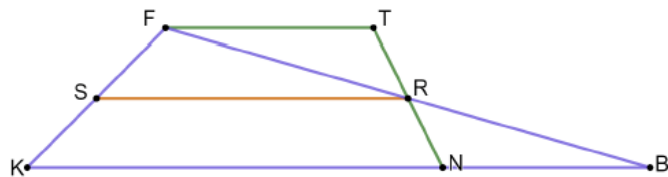
1. $\triangle ACN$ is shown with midsegment $\overline{KF}$.



Fill in the blanks to complete the triangle midsegment theorem.

The triangle midsegment theorem states that the midsegment of a triangle is **parallel** to the third side it does not touch and is **half** the length of the third side.

2. $\triangle FBK$ with midsegment $\overline{SR}$ and trapezoid $FTNK$ are shown where $\overline{FT} \parallel \overline{KN}$.



- a. Explain why $SF = KS$.
 $\overline{SR}$ is the midsegment of $\triangle FBK$, so point S is the midpoint of $\overline{FK}$. By definition of a midpoint, $SF = KS$.



Students are introduced to the midsegment of a triangle. The triangle midsegment theorem will be explored more in a later unit. During this lesson, students just need a surface understanding that the midsegment exists, and its relationship to the base of the triangle.



Before moving into question 2, lead students in recalling the properties of a trapezoid and what defines a quadrilateral as a trapezoid.

8.7.3 Guided Instruction: Trapezoid Midsegment Theorem (continued)

- b. If $\overline{NB} \cong \overline{FT}$, justify why $\overline{NR} \cong \overline{RT}$.
 $\angle FTR \cong \angle BNR$ because when parallel lines are crossed by a transversal alternate interior angles are congruent. $\angle TRF \cong \angle NRB$ because vertical angles are congruent. $\triangle FTR \cong \triangle BNR$ by AAS congruence. $\overline{NR} \cong \overline{RT}$ because of CPCTC.
- c. Discuss with your partner what information is necessary so that the definition of a trapezoid can be used to state that $FTRS$ is a trapezoid. Summarize your discussion.
At least one pair of sides must be parallel.
- d. Discuss with your partner what else needs to be determined so that it can be stated that $\overline{SR}$ is a midsegment of $FKNK$. Summarize your discussion.
 S is the midpoint of $\overline{KF}$ and R is the midpoint of $\overline{TN}$



Trapezoid Midsegment Theorem

The midsegment of a trapezoid is parallel to the bases and is half the sum of the bases.

8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

Component Overview: Students apply Trapezoid Midsegment Theorem to find missing lengths.



8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

1. Trapezoid $WDLY$ with midsegment $\overline{FZ}$ is shown.



- a. If $WD = 14.2$ and $YL = 29.7$, what is the length of $\overline{FZ}$?

$$FZ = \frac{1}{2}(WD + YL) = \frac{1}{2}(14.2 + 29.7) = 21.95$$

- b. If $WDLY$ is an isosceles trapezoid and $m\angle DLY = 53^\circ$, what is $m\angle WFZ$?

$$m\angle WFZ = 53^\circ$$

- c. If $FZ = 11.4$ and $YL = 17.6$, what is the length of $\overline{WD}$?

$$FZ = \frac{1}{2}(WD + YL)$$

$$11.4 = \frac{1}{2}(WD + 17.6)$$

$$WD = 5.2$$

- d. What is the value of k if $WD = 4k + 4.2$, $FZ = 7k + 2.1$, and $YL = 6k + 16.8$?

$$FZ = \frac{1}{2}(WD + YL)$$

$$7k + 2.1 = \frac{1}{2}((4k + 4.2) + (6k + 16.8))$$

$$k = 4.2$$

- e. If $WF = \frac{2}{3}p$ and $WY = 12 - 2p$, what is the length of $\overline{WY}$?

$$2(WF) = WY$$

$$2\left(\frac{2}{3}p\right) = 12 - 2p$$

$$p = 3.6$$

$$WY = 12 - 2(3.6) = 4.8$$



Each part of question 1 is independent (question 1 parts a – e can be completed independently). Consider posting the questions around the room for a mini Gallery Walk. Students can be broken into groups to work on a question and then rotate to check the work and make comments on the other questions.



MA.K12.MTR.3.1: Mathematical Fluency

Students need the opportunity to develop their own methods and explore other ways of thinking to build flexibility in fluency.

- Why did you take that approach?
- What would happen if you tried this as your first step?
- Could there be other answers?
- How do you know that (pick a number) is not another possible answer?
- What does this tell you about the trapezoid?

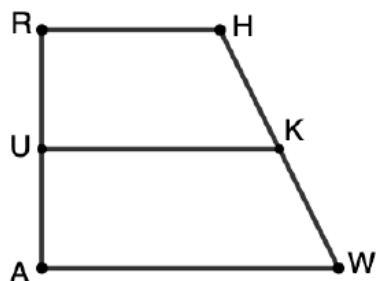
8.7.5 Your Turn!

Component Overview: Students engage in independent practice, applying properties of trapezoids to solve problems.



8.7.5 Your Turn!

1. Trapezoid $RHWA$ is shown with midsegment $\overline{UK}$. Find the values of x and y if $RU = -2x - 2y$, $AU = 10$, $RH = 4x$, $UK = 14$, and $AW = -y + 8$.



$$RU = AU$$

$$-2x - 2y = 10$$

$$UK = \frac{1}{2}(RH + AW)$$

$$14 = \frac{1}{2}(4x + (-y + 8))$$

$$10 = 2x - \frac{1}{2}y$$

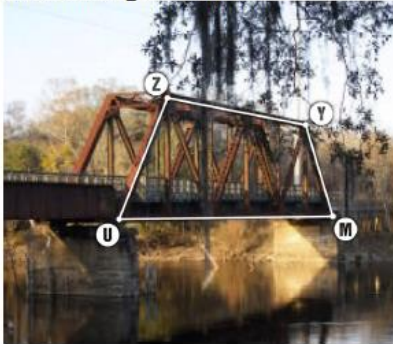
$$x = 3, y = -8$$



Question 1 involves solving a system of equations. Students may need a brief review of how to solve a system of equations in two variables.

8.7.5 Your Turn! (continued)

2. Located on the Nature Coast State Trail in north-central Florida is the Suwannee River Trestle, a bridge that is shaped like an isosceles trapezoid. The state has been inspecting older bridges to ensure their safety and has determined that the Suwannee River Trestle, represented as trapezoid $ZYMU$, needs an additional support brace through its middle.



The support brace is to connect sides $\overline{ZU}$ and $\overline{YM}$ at their midpoints. If $ZU = 28$ feet (ft) and the perimeter of the trestle measures 153 ft, what is the length, in ft, of the new support brace?

$$\begin{aligned}\text{Perimeter} &= ZU + YM + ZY + UM \\ 153 &= 28 + 28 + ZY + UM \\ 97 &= ZY + UM\end{aligned}$$

$$\begin{aligned}\text{Midsegment} &= \frac{1}{2}(ZY + UM) \\ \text{Midsegment} &= \frac{1}{2}(97) \\ \text{Midsegment} &= 48.5 \text{ ft}\end{aligned}$$



Three Reads

Consider prompting students to engage in Three Reads if they are struggling to decode the word problem on their own.



What information in the question 2 stem is needed to solve this problem?

8.7.6 Wrap-Up: Cool Down

Component Overview: Students capture their learning by writing a note to an absent classmate.



8.7.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand the trapezoid midsegment theorem.

Answers will vary. The midsegment of a trapezoid is the line segment that connects the midpoints of the two legs of the trapezoid. A midsegment is parallel to the bases of the trapezoid and its length is equal to half the sum of the length of the two bases.



Use the summaries written here to determine understanding and remediation needs for this lesson.